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Growth of Gold Bipyramids with Improved Yield and Their Curvature-Directed Oxidation

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The surface-plasmon properties of Au bipyramids are investigated using the finite-difference time-domain method. It is found that both the extinction cross sections and local electric-field enhancements of Au bipyramids are larger than those of Au nanorods that have longitudinal surface plasmon (LSP) wavelengths close to those of Au bipyramids. Following this result the growth of Au bipyramids using cationic surfactants of variously sized headgroups as stabilizing agents is carried out. It is found that the growth using cetyltributylammonium bromide (CTBAB) produces Au bipyramids with tunable LSP wavelengths in high yields. The oxidation behaviour of Au bipyramids using hydrogen peroxide as the oxidizing agent is fully explored and the oxidation is found to occur preferentially at highly curved surface sites. It is further demonstrated that the oxidation rate can be controlled by varying the amounts of hydrogen peroxide and hydrochloric acid. This oxidation approach can be used in conjunction with the seed-mediated growth in CTBAB solutions to produce Au bipyramids, the LSP wavelengths of which are finely tunable from 650 to 1300 nm.

Keywords:

- bipyramids
- gold
- oxidation
- surface plasmon resonance
- surfactants

1. Introduction

Gold nanorods exhibit strong optical extinction at their longitudinal surface plasmon (LSP) wavelengths, which are tunable from the visible to near-infrared region by varying their aspect ratios. They have been demonstrated to func-

tion as scattering^[1,2] and two-photon-absorption^[3] chromophores for biological imaging of blood vessels and cancer cells, biological sensors,^[4,5] carriers for drug delivery,^[6] and agents for photothermal cancer therapy.^[2] They have been incorporated into polymers to create hybrid optical materials^[7] and deposited onto bacteria to create bioelectronic materials.^[8] They have been assembled onto flat surfaces to function as substrates for surface-enhanced Raman spectroscopy (SERS).^[9] In addition, interesting chain^[5,10,11] and ring structures^[12] have been formed from Au nanorods using templates or various thiol molecules.

The electric field in the region close to the surfaces of noble metal nanostructures is enhanced relative to the far field when they are illuminated at their surface-plasmon wavelengths. For Au nanorods, such local field enhancements, as well as their extinction cross sections and surface-plasmon wavelengths, are essential for realizing their poten-

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tial optical and biotechnological applications. In general, large field enhancements are beneficial to SERS and two-photon luminescence imaging, and high extinction cross sections are favourable for scattering-based imaging. Therefore, Au nanostructures that exhibit both larger local field enhancements and extinction cross sections than Au nanorods and also possess tunable surface-plasmon wavelengths similar to Au nanorods are highly desired.

Gold bipyramids have a pentagonal base and two sharp apexes. Similar to Au nanorods, bipyramids also possess two surface-plasmon resonances corresponding to the electron oscillations along the longitudinal and transverse directions. The LSP wavelengths can be tuned systematically by varying the ratio of the height to the base-edge length. Because the apexes of Au bipyramids are sharper than the ends of nanorods, the local electric-field enhancements associated with bipyramids should be larger than that associated with nanorods, according to the lightning-rod effect.^[13] Au bipyramids should thus be more advantageous than Au nanorods for two-photon luminescence imaging and SERS.

A seed-mediated method has previously been used to grow Au bipyramids in aqueous solutions using cationic surfactants as stabilizing agents in the presence of AgNO_3 .^[14,15] The growth using citrate-stabilized Au nanoparticle seeds in cetyltrimethylammonium bromide (CTAB) solutions produces Au bipyramids with LSP wavelengths in the range of 750–850 nm and yields of $\approx 30\%$. Cetyltriethylammonium bromide (CTEAB) has also been used as a stabilizing agent to grow Au bipyramids.^[16] The LSP wavelengths of the resulting bipyramids can be varied from 700 to 1100 nm by changing the amount of seeds relative to that of the Au precursor. However, the maximum yields of bipyramids are below 30%, and the yields decrease rapidly for bipyramids with longer LSP wavelengths.

We have previously used cetyltripropylammonium bromide (CTPAB) and cetyltributylammonium bromide (CTBAB) surfactants to grow Au nanorods using CTAB-stabilized Au nanoparticle seeds through the seed-mediated method.^[17] These surfactants have larger headgroups than CTAB and CTEAB. A systematic comparison of the nanorod growth using these four surfactants indicates that the average aspect ratios of the Au nanorods increase and the nanorod growth rates decrease as the surfactant headgroup gets larger. The reason for this headgroup size-dependent nanorod-growth behaviour has been postulated to be that the bilayers formed on the surfaces of Au nanorods from the surfactants with larger headgroups are more stable. On the basis of these previous observations, we expect that the seed-mediated growth using CTPAB and CTBAB as stabilizing agents might produce Au bipyramids with LSP wavelengths tunable over a broader spectral range and in improved yields.

In this paper, we first present the results of finite-difference time-domain (FDTD) calculations performed on Au bipyramids and nanorods that have similar LSP wavelengths. We then describe the growth of Au bipyramids in CTBAB solutions using citrate-stabilized Au nanoparticle seeds through the seed-mediated method. Finally, curvature-directed oxidation of Au bipyramids using hydrogen perox-

ide as the oxidizing agent is demonstrated. The oxidation rate is shown to be dependent on the amount of the oxidizing agent, the acid concentration, the stabilizing surfactant, and the type of Au nanostructures. This oxidation approach allows us to produce Au bipyramids with their LSP wavelengths finely controlled between 650 and 1300 nm.

2. Results and Discussion

2.1. FDTD Calculation

The surface-plasmon properties of Au bipyramids and nanorods were investigated and compared using the FDTD method. The FDTD method can be used to calculate the optical response of noble metal nanostructures of arbitrary geometries to an incident light wave.^[18] Surface-plasmon properties such as absorption and scattering cross sections as well as local electric-field enhancements can be obtained from this method. Such properties are generally difficult to measure experimentally but they are very important for the creation of optical and biotechnological materials and devices using noble metal nanostructures.

Gold bipyramids and nanorods of representative sizes that are readily available from the seed-mediated growth were chosen for FDTD calculation. Their transmission electron microscopy (TEM) images are shown in Figure 1 A and B. The average height and base width of the bipyramids

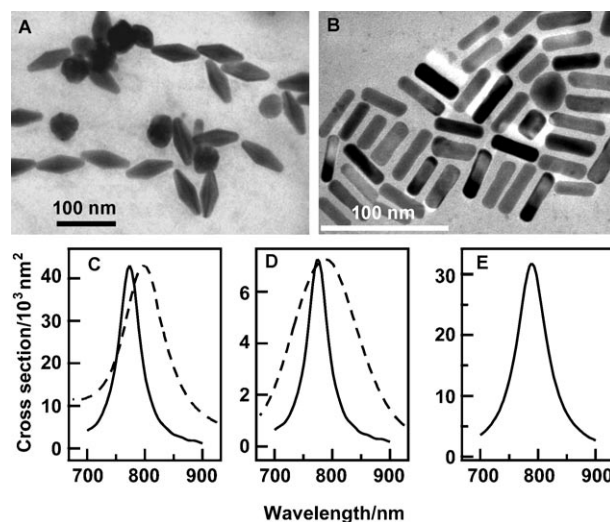


Figure 1. FDTD calculation on averagely sized Au bipyramids and nanorods. A) TEM image of Au bipyramids. B) TEM image of Au nanorods. C) Extinction spectrum taken from the Au bipyramid solution (dashed line) and that obtained from FDTD calculation on an averagely sized bipyramid (solid line). D) Extinction spectrum taken from the Au nanorod solution (dashed line) and that obtained from FDTD calculation on an averagely sized nanorod (solid line). E) Extinction spectrum obtained from FDTD calculation on a nanorod with its volume approximately equal to that of the averagely sized bipyramid. Only the extinction peak corresponding to the LSP resonance is shown. The experimentally measured spectra were scaled against the calculated spectra. The bipyramids were grown in a CTEAB solution, and the nanorods were grown in a CTAB solution.

measured from their TEM images are 88 ± 3 nm and 32 ± 1 nm, respectively. Their apexes are rounded and the average apex angle formed between the extended side edge lines is $30^\circ \pm 2^\circ$. The bipyramid ensemble sample has a LSP extinction peak at 798 nm with a full width at half-maximum (FWHM) of 82 nm (dashed line in Figure 1C). The nanorods have an average length and diameter of 41 ± 4 and 13 ± 2 nm, respectively. The nanorod ensemble sample has a LSP extinction peak at 787 nm with a FWHM of 137 nm (dashed line in Figure 1D). The LSP extinction peak of the bipyramid sample has a smaller FWHM because the size and shape distributions of the bipyramids are narrower than those of the nanorods.

For FDTD calculation, the sizes of the bipyramid and nanorod were taken to be as close as possible to their respective average values obtained from TEM imaging. Specifically, the height and base edge length of the bipyramid are 87 and 20.4 nm, respectively, and its apex angle is 30° . Its apexes were modelled as spherical caps with five cut facets. The diameter of the capped sphere is 14.1 nm, which is determined by the height and apex angle of the bipyramid. The length and diameter of the nanorod are 42 and 12 nm, respectively. Its ends are modelled as half spheres. Figure 1C and D show the extinction curves obtained from the FDTD calculation at a mesh size of 0.5 nm for the bipyramid and nanorod, respectively (see also Supporting Information (SI), Figure S1 and S2). The extinction spectrum of the bipyramid in the region of the longitudinal surface-plasmon resonance is similar to that of the nanorod. The calculated LSP peak wavelengths at 773 and 775 nm (Table 1) are both blue-shifted relative to the ensemble values for the bipyramid and nanorod, respectively. This is because, in general, larger bipyramids and nanorods have larger extinction cross sections. They contribute more to the extinction spectrum than smaller ones during ensemble measurements. On the other hand, the average sizes of the bipyramid and nanorod used for the FDTD calculation are determined from TEM imaging, where particles of different sizes are sampled at an equal probability. This causes a blue-shift in the FDTD-calculated LSP relative to that obtained from ensemble measurements for both the bipyramid and nanorod. Moreover, the FWHMs obtained from the FDTD calculation are 47 nm (97 meV) and 45 nm (93 meV) for the bipyramid and nanorod, respectively (Table 1). For comparison, the homogeneous FWHMs obtained previously by measuring the white-light scattering spectra from individual bipyramids^[15] and nanorods^[19] of similar LSP wavelengths are 100 and 80 meV, respectively. The FWHMs measured from the

ensemble samples are larger than the homogeneous FWHMs obtained from individual bipyramids and nanorods, suggesting that the ensemble bipyramid and nanorod samples have an inhomogeneous size and shape distribution.

A marked difference between the bipyramid and nanorod is that the extinction cross section of the bipyramid is about 6 times that of the nanorod at their LSP wavelengths (Table 1). Au nanorods have previously been treated approximately as ellipsoids and Gans' theory has been used to analyze their surface plasmon properties.^[20] According to Gans' theory, the extinction cross section of an ellipsoid is proportional to its volume. The volume of the bipyramid used for FDTD calculation is about 6 times that of the nanorod. Therefore, the difference in the extinction cross section between the bipyramid and nanorod obtained from FDTD calculation is in agreement with that predicated by Gans' theory.

The local electric-field-intensity enhancement contours associated with the LSP resonance are shown in Figure 2 and SI Figure S3 for the averagely sized Au bipyramid and nanorod. The propagation direction of the incident light wave is perpendicular to the longitudinal direction and its polarization is parallel to the longitudinal direction. It is seen that the bipyramid supports a very strong local field enhancement at its apexes, with a maximal intensity enhancement value of 6100. In comparison, the maximal field intensity enhancement of the nanorod at its ends is 1700 (Table 1). The field intensity enhancement supported by the bipyramid is 3.6 times that by the nanorod. The field enhancement exhibited by the bipyramid is comparable to that reported recently for hematite-Au core/shell prolate spheroidal nanostructures^[21] and is several times larger than those reported previously for bowtie junctions fabricated using electron beam lithography.^[22] Such strong local field enhancements make Au bipyramids highly attractive for use in designing substrates for SERS and creating hybrid materials with enhanced nonlinear optical properties.

We further performed FDTD calculations on a nanorod whose volume is approximately equal to that of the averagely sized Au bipyramid in order to investigate if the particle volume has an effect on the local electric-field enhancement. The length and diameter of the Au nanorod were chosen to be 78 and 22 nm, respectively, so that its volume and LSP peak wavelength are close to those of the bipyramid (Figure 1E, SI Figure S4, and Table 1). It is seen that the maximal field intensity enhancement of this larger nanorod is about one sixth that of the bipyramid, indicating that the maximal field enhancement is mainly determined by the

Table 1. FDTD calculation results.

Sample	Bipyramid	Small nanorod	Large nanorod
Volume [nm ³]	27 000	4300	27 000
Wavelength [nm] ^[a]	773	775	789
FWHM [nm] ^[b]	47	45	61
Cross section [nm ²] ^[c]	43 000	7300	32 000
Intensity enhancement ^[d]	6100	1700	1000

[a] The LSP peak wavelength. [b] The LSP peak linewidth. [c] The extinction cross section at the LSP peak. [d] The maximal electric-field-intensity enhancement associated with the LSP resonance.

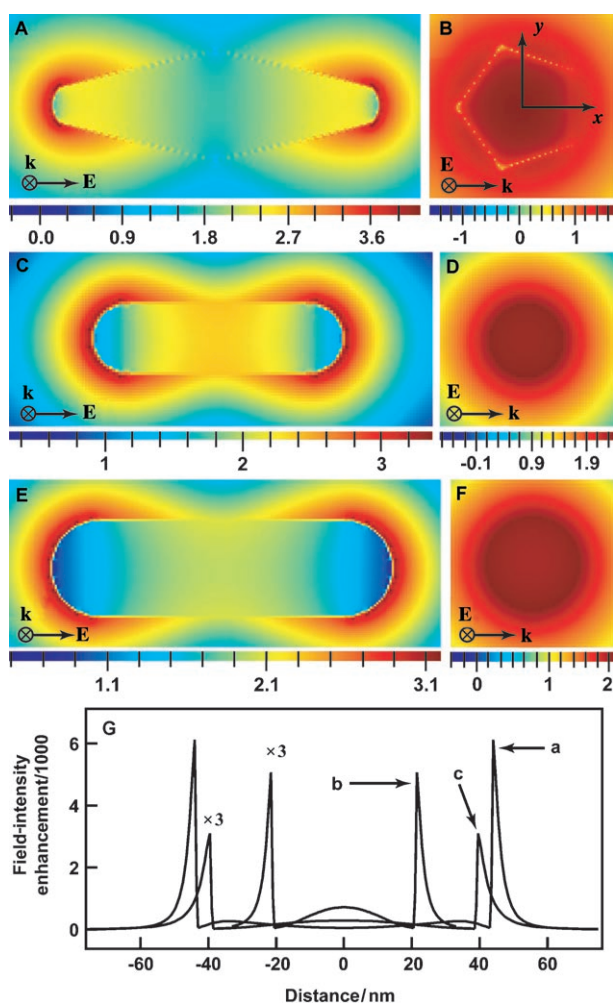


Figure 2. Electric-field-intensity enhancement contours at the logarithmic scale obtained from FDTD calculation at a mesh size of 0.5 nm. A, B) y - z and x - y planes of the averagely sized bipyramid. C, D) The averagely sized nanorod. E, F) The nanorod that has a volume approximately equal to that of the bipyramid. G) Electric-field-intensity enhancement profiles along the longitudinal direction for a) the bipyramid, b) the nanorod, and c) the nanorod with its volume approximately equal to that of the bipyramid. The origin is at the centers of the bipyramid and both nanorods. They are excited at their respective LSP peak wavelengths. The bipyramid is 87 nm in height, 20.4 nm in base edge length, and its apex angle is 30° . Its apexes are modeled as spherical caps 14.1 nm in diameter with five cut facets. The smaller nanorod is 42 nm in length and 12 nm in diameter. The larger nanorod is 78 nm in length and 22 nm in diameter.

nanoparticle shape. In addition, the extinction cross section of the nanorod is smaller than that of the bipyramid at their LSP peak wavelengths, although their volumes are roughly the same, suggesting that the particle shape also has an effect on the extinction cross section.

2.2. Growth of Gold Bipyramids

It is seen from FDTD calculations that Au bipyramids exhibit larger electric-field enhancements and extinction

cross sections than nanorods at similar LSP wavelengths. Au bipyramids are therefore more favourable for optical applications where large extinction cross sections and high electric-field enhancements are desired. Au bipyramids have previously been grown in CTAB and CTEAB solutions using a seed-mediated method.^[14–16] However, the yields of bipyramids with LSP wavelengths in the range of 750–850 nm are limited below $\approx 30\%$, and the yields decrease rapidly for bipyramids with longer LSP wavelengths.

Hydrochloric acid has been used during the growth of Au bipyramids in CTAB solutions with ascorbic acid as a reducing agent.^[15] The presence of hydrochloric acid can slow down the growth and increase the yield of Au bipyramids due to the decrease in the reducing capability of ascorbic acid. On the other hand, it has been found that the use of cationic surfactants with increasingly larger headgroups can slow down the growth of Au nanorods.^[17] The binding of surfactants to the surfaces of Au nanorods gets stronger as the headgroups become larger, probably due to reduced electrostatic repulsion among the headgroups. We therefore carried out the growth of Au bipyramids using CTBAB as a stabilizing agent. Figure 3 shows the extinction spectra of

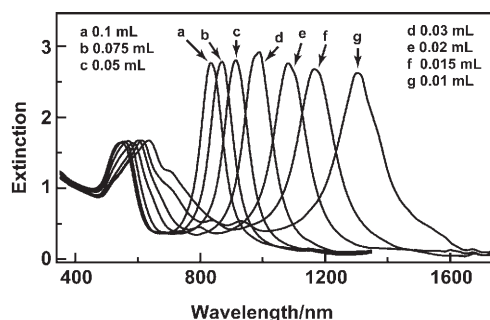


Figure 3. Extinction spectra of Au bipyramids grown in CTBAB solutions with varying volumes of the citrate-stabilized seed solution. The growth solutions are composed of HAuCl_4 (0.01 M, 0.3 mL), ascorbic acid (0.1 M, 0.1 mL), AgNO_3 (0.01 M, 0.45 mL), and CTBAB (0.01 M, 7.125 mL).

the Au bipyramids grown in CTBAB solutions with varying volumes of the citrate-stabilized seed solution. As the volume of the seed solution is reduced, the LSP peak of the Au bipyramids shifts to the longer wavelength. A tunable range of 800–1300 nm in the LSP wavelength can be readily achieved during our growth. More remarkably, as the LSP peak wavelength becomes longer, the extinction values at both the longer-wavelength and shorter-wavelength peaks remain almost constant around 2.8 and 1.6, respectively. This suggests that the yields of the Au bipyramids remain approximately constant as they grow larger because the longer-wavelength extinction peak is solely due to the LSP resonance of the bipyramids and the shorter-wavelength peak is contributed by both the transverse plasmon resonance of the bipyramids and the plasmon resonance of the roughly spherical Au nanoparticles that were grown together with the bipyramids. In addition, the FWHMs of the LSP peaks of the bipyramids are in the range of 80–180 nm,

while those of the Au nanorods that were grown under similar conditions with similar LSP wavelengths are in the range of 150–310 nm.^[16,17] The difference in the line width of the LSP peak suggests that the size and shape of the Au bipyramids are more uniform than those of the Au nanorods.

TEM imaging was performed to characterize the sizes and yields of the Au bipyramids grown in CTBAB solutions. Figure 4A–D show four representative TEM images of the growth products obtained with different volumes of the seed solution. The growth products are mainly composed of bipyramids and faceted spherical nanoparticles. The yields of bipyramids were found to be 50–60% for all of these

samples by counting 1000–1500 particles on their TEM images. This is consistent with the observation that their LSP peak extinction values remain almost constant as their LSP peak wavelengths become longer. The sizes of these bipyramids are relatively uniform. We measured the heights and base widths of about 200 bipyramids per sample and obtained the average. Figure 4E and F show the changes of the average base width and height as a function of the added volume of the seed solution during growth. As the volume of the seed solution is reduced, both the average base width and height are increased. The average aspect ratio of the height to the base width is also increased (Figure 4G), suggesting that the increase in the height is faster than that in the base width. Figure 4H further shows the relationship between the LSP peak wavelength and the average aspect ratio. The LSP wavelength increases nonlinearly as the aspect ratio gets larger. The relationship between the LSP wavelength and aspect ratio observed here for Au bipyramids is different from that for Au nanorods. The LSP wavelength of Au nanorods increases linearly with increasing aspect ratio, as shown in previous experiments.^[23,24] We believe that the difference in the relationship between the LSP wavelength and aspect ratio observed for Au bipyramids and nanorods can be ascribed to the effect of the particle volume on the LSP wavelength. Recent calculations using the discrete dipole approximation method have indicated that the LSP wavelength of Au nanorods increases linearly as a function of the aspect ratio at fixed particle volumes and that it increases significantly at fixed aspect ratios when the particle volume is increased.^[25] For Au bipyramids, the particle volume is seen to increase significantly and nonlinearly from 1×10^4 to $12 \times 10^4 \text{ nm}^3$ as the aspect ratio is increased from 3.4 to 5.5. The shape of the curve appears similar to that for the LSP wavelength

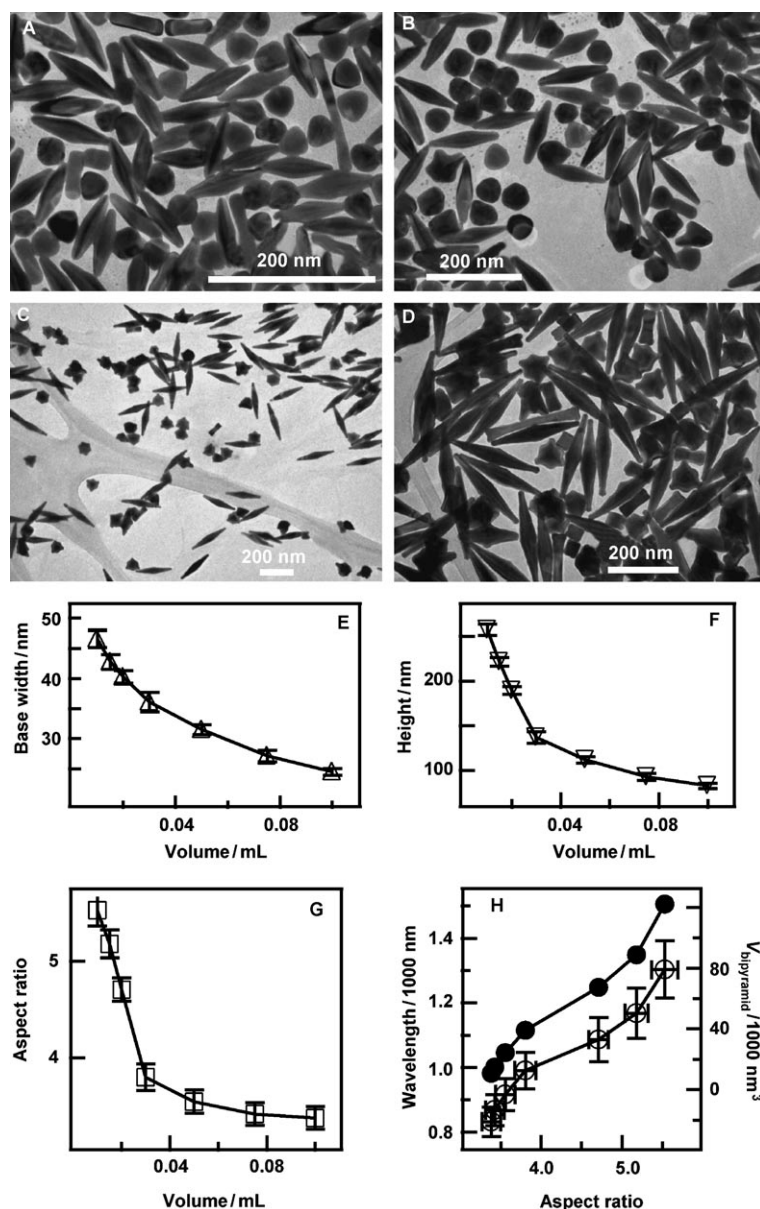


Figure 4. A–D) TEM images of the Au bipyramid samples whose extinction spectra correspond to the curves b, d, f, and g shown in Figure 3, respectively. E–G) Base widths, heights, and aspect ratios, respectively, of the Au bipyramids as a function of the volume of the seed solution used for the growth. H) LSP peak wavelength (left axis, hollow circles) and bipyramid volume (right axis, solid circles) changes as a function of the aspect ratio of the bipyramids. The bipyramid volume is calculated without the consideration of the blunted apexes.

versus the aspect ratio (Figure 4H). In contrast, the particle volumes of Au nanorods grown using the seed-mediated method in previous experiments are estimated to vary within a much smaller range of 0.2×10^4 to 1.5×10^4 nm³ as the aspect ratio is changed in the range of 2.5–8.5.^[23,24] Therefore the LSP wavelength of Au bipyramids exhibits a nonlinear change as a function of the aspect ratio while that of Au nanorods appears linear.

Ascorbic acid was used as the reducing agent for the seed-mediated growth of Au bipyramids. The stoichiometric molar ratio of ascorbic acid to HAuCl₄ is 1.5 according to the redox reaction.^[26] However, it has been found in our experiments that a molar ratio of ascorbic acid to HAuCl₄ above 3.0 is generally beneficial to the growth of Au bipyramids with tunable LSP wavelengths in high yields. If the molar ratio of ascorbic acid to HAuCl₄ is reduced to be close to the stoichiometric value, the yields of bipyramids with longer LSP wavelengths decrease considerably (SI Figure S5).

We further carried out the seed-mediated growth experiments using CTAB, CTEAB, and CTPAB surfactants as stabilizing agents (SI Figure S6–8). By taking together the growth results in the solutions of the four surfactants, it is seen that the yields of Au bipyramids generally increase as the surfactant headgroup becomes larger. CTBAB has the largest headgroup, and the yield of Au bipyramids grown in CTBAB solutions is the highest among the four surfactants. This is exactly what we have expected. During growth, surfactants form a bilayer on the surfaces of bipyramids.^[27] The positively charged headgroups in the inner layer are bound to the surfaces through electrostatic interactions. The increase in the yields of Au bipyramids probably results from the stronger binding of the surfactants to the surfaces of bipyramids as their headgroups get larger.^[17]

2.3. Oxidation of Gold Bipyramids

The ability to tailor the shapes and sizes of inorganic nanostructures is critical to modern materials chemistry because the intrinsic properties of most nanostructures are strongly dependent on their shapes and sizes. A very popular approach that has been used is the formation of nuclei from atoms and molecules at the first stage, followed by growth into variously shaped nanostructures at the second stage through careful control of the growth rates along different lattice directions. In addition to this general growth approach, reshaping and resizing through chemical reactions can also be used to make inorganic nanostructures of various shapes and sizes. For example, it has been demonstrated that Au nanorods can be shortened to produce nanorods of smaller aspect ratios through anisotropic oxidation reactions.^[24,28] One advantage of the shortening approach to the preparation of Au nanorods is that the shortening process can be monitored in real time and nanorods of any aspect ratio smaller than that of starting nanorods can be obtained.

We carried out the oxidation experiments on the Au bipyramids grown in CTBAB solutions in order to obtain bipyramids of finely tuned LSP wavelengths and compare the

oxidation behaviour of bipyramids with that of nanorods. Cyanide, persulfate, Au^{III}, and oxygen have been used in previous experiments to oxidize Au nanorods.^[24,28] H₂O₂ was chosen in our experiments to oxidize Au bipyramids because it is nontoxic and easy to handle. Figure 5A and B

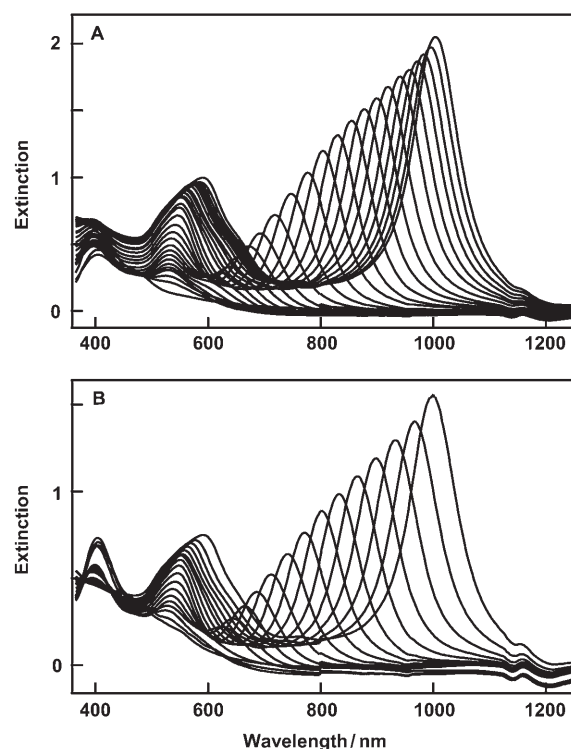


Figure 5. Extinction spectra taken as a function of time during the oxidation of Au bipyramids. The volume ratios of H₂O₂ (35 wt. %) to as-grown Au bipyramid solution are A) 0.5 and B) 1.0. Au bipyramids were grown in CTBAB solutions. The spectra were taken every 30 min after the addition of H₂O₂.

show the extinction spectra taken as a function of time for the oxidation using two different amounts of H₂O₂. The LSP and transverse plasmon peak wavelengths of the starting Au bipyramids are around 1000 and 590 nm, respectively. During oxidation, the LSP peak exhibits a significant blue-shift, while the transverse plasmon peak only blue-shifts slightly. Both peaks decrease in intensity. With continuing oxidation, the LSP peak becomes a shoulder of the transverse plasmon peak and finally disappears, suggesting the transformation of bipyramids into nanospheres. Au nanospheres are eventually completely oxidized, as indicated by the disappearance of the single extinction peak around 530 nm. Moreover, a weak peak appears around 400 nm with oxidation, and it becomes stronger and stronger throughout the oxidation process. This peak results from the metal–ligand charge-transfer absorption of HAuBr₄, indicating the oxidation of Au⁰ to Au^{III}.

The extinction spectra shown in Figure 5A and B were taken every 30 min. It is seen that the blue-shift in the LSP wavelength is faster as the amount of H₂O₂ used for oxidation is increased. To quantify the oxidation rate, we plotted

the LSP wavelength as a function of time and fitted the data points using a line (SI Figure S9). The resulting slope was used to characterize the oxidation rate of Au bipyramids under different conditions. The oxidation rates are determined to be 0.72 and 1.01 nm min⁻¹ (Table 2) when the volume ratios of H₂O₂ to as-grown Au bipyramid solution are 0.5 and 1.0, respectively.

Table 2. Oxidation rates of gold bipyramids and nanorods under different conditions.

Oxidation condition ^[a]	Oxidation rate [nm min ⁻¹] ^[b]
2 mL Au BPs in CTBAB, 1 mL H ₂ O ₂	0.72 (0.02)
1 mL Au BPs in CTBAB, 1 mL H ₂ O ₂	1.01 (0.01)
2 mL Au BPs in CTBAB, 1 mL H ₂ O ₂ , 0.02 mL HCl	3.7 (0.1)
1 mL Au BPs in CTBAB, 1 mL H ₂ O ₂ , 0.01 mL HCl	4.4 (0.9)
2 mL Au NRs in CTBAB, 1 mL H ₂ O ₂	1.4 (0.2)
1 mL Au NRs in CTBAB, 1 mL H ₂ O ₂	3.2 (0.3)
2 mL Au NRs in CTAB, 1 mL H ₂ O ₂	15.3 (0.8)
1 mL Au NRs in CTAB, 1 mL H ₂ O ₂	40 (9)

[a] BPs: bipyramids. NRs: nanorods. As-grown Au bipyramid and nanorod solutions were used for oxidation. The concentrations of CTBAB and CTAB are 0.01 and 0.1 M, respectively. Those of H₂O₂ and HCl are 35 wt.% and 1.0 M, respectively. [b] The numbers in parentheses are standard deviations.

The oxidation rate of Au bipyramids can not only be changed by varying the amount of H₂O₂, but also by adding HCl. Figure 6A and B show the evolutionary extinction spectra acquired for the oxidation with the addition of HCl. The pH of as-grown Au bipyramid solutions is 3–4. The oxidation rate is increased several times if the solution pH is adjusted to ≈2 by the addition of a small amount of HCl (Table 2). The faster oxidation of Au bipyramids obtained by the addition of HCl is because the oxidizing capability of H₂O₂ is enhanced by the increase in the acid concentration.

Our demonstrated ability to control the oxidation rate is very useful for the production of Au bipyramids with finely tuned LSP wavelengths. It has been shown above that the LSP wavelengths of Au bipyramids can also be tailored during growth by varying the amount of seeds relative to that of the Au precursor. However, we have found that the LSP wavelengths of Au bipyramids that are grown following the same procedure often vary from batch to batch due to unavoidable fluctuations in growth conditions. In addition, the tuning of the LSP wavelength by varying the amount of seed is discrete in nature, while that through oxidation is continuous. A combination of these two approaches makes possible the preparation of Au bipyramids with LSP wavelengths controlled precisely over a wide spectral range from 650 to 1300 nm.

Au bipyramids at varying stages of oxidation can be obtained by centrifugation and redispersion into aqueous solutions if the oxidation rate is sufficiently small. The resulting samples exhibit vivid colors (SI Figure S10). The evolution of their morphologies and sizes was monitored using TEM (Figure 7A–F). Oxidation at the early stage mainly occurs

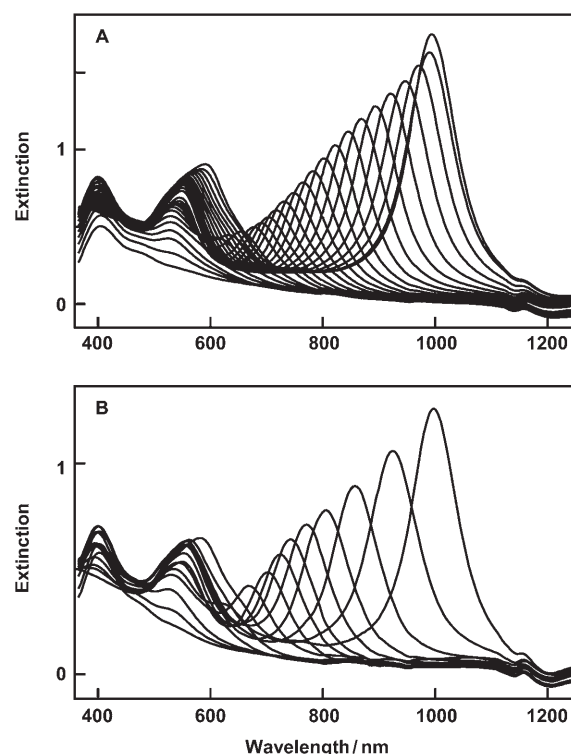


Figure 6. The effect of HCl on the oxidation rate of Au bipyramids grown in CTBAB solutions. A) The mixture solution for oxidation is composed of 2 mL as-grown Au bipyramid solution, 1 mL H₂O₂ (35 wt. %), and 0.02 mL of HCl (1.0 M). The spectra were acquired every 5 min. B) The mixture solution for oxidation is composed of 1 mL as-grown Au bipyramid solution, 1 mL H₂O₂ (35 wt. %), and 0.01 mL of HCl (1.0 M). The spectra from right to left were acquired 1 min, 3 min, 5 min, 7 min, 9 min, 12 min, 15 min, 20 min, 30 min, 45 min, 1 h, 1.5 h, 2 h, and 2.5 h consecutively after the addition of H₂O₂ and HCl.

at the apexes of bipyramids. The apexes become blunter and the bipyramids are shortened (Figure 7A). With continuing oxidation, the apexes of the bipyramids are further blunted. The vertexes at the base and the side edges are gradually smoothed (Figure 7B and C). As a result, the bipyramids become ellipsoidal. After this stage, the ellipsoids shrink in both their major and minor axes (Figure 7D and E). Finally they become spheres (Figure 7F). It is clear from TEM imaging that the oxidation reaction preferentially occurs at surface sites with high curvature, so that sharp apexes and edges gradually disappear and smoother surfaces are produced. Such curvature-directed oxidation has also been observed previously during the oxidation of Au nanorods using Au^{III} and oxygen, where the length of nanorods is reduced gradually while the diameter remains almost unchanged before nanorods are transformed into nanospheres.^[24]

The sizes of the products obtained from the oxidation of Au bipyramids were measured on TEM images. The width was measured at the middle since it varies along the longitudinal direction. The average middle widths and heights obtained from the measurements of ≈200 particles per sample are shown in Figure 7G and H as a function of the oxida-

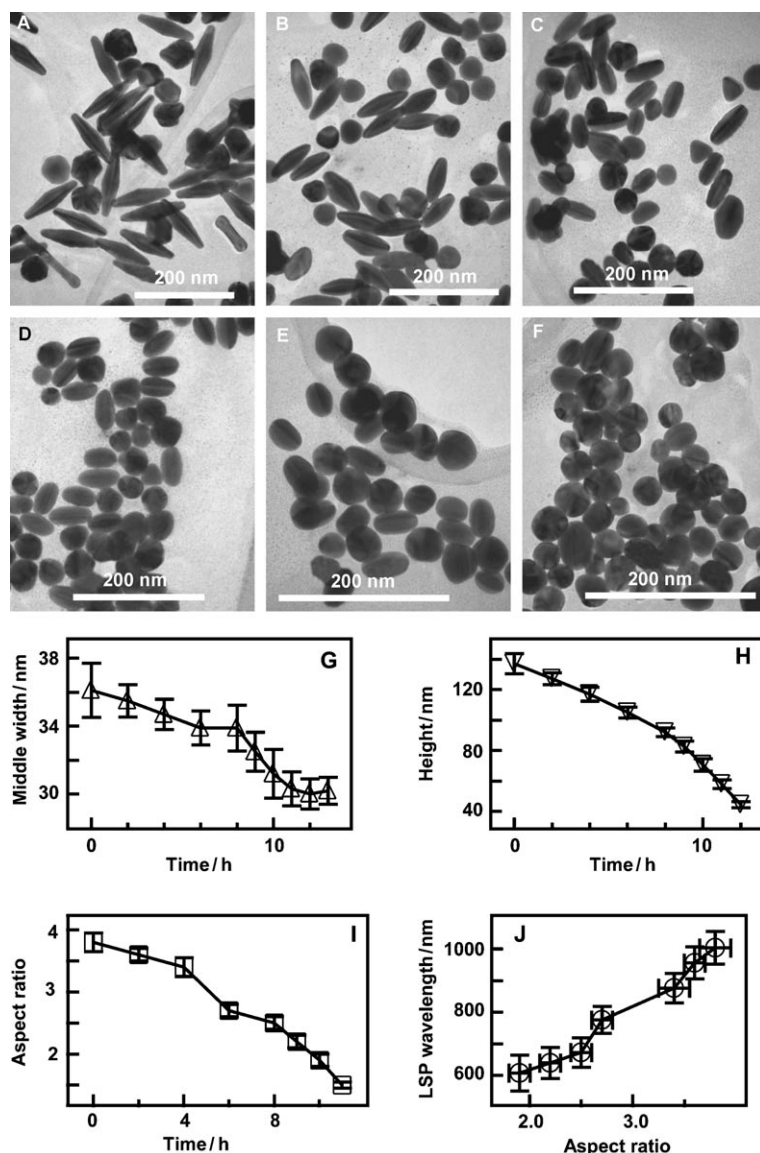


Figure 7. A–F) TEM images of the products obtained 4 h, 8 h, 10 h, 11 h, 12 h, and 13 h after the addition of H_2O_2 into as-grown Au bipyramid solution for oxidation. The volume ratio of H_2O_2 to the bipyramid solution is 0.5. G–I) Middle widths, heights, and aspect ratios, respectively, of the products as a function of the oxidation time. J) The relationship between the LSP peak wavelength and aspect ratio of the products. The starting bipyramid sample corresponds to that shown by curve d in Figure 3.

tion time. Both become smaller with increasing extent of oxidation. The average aspect ratio of the height to the middle width also decreases with increasing oxidation time (Figure 7I), indicating that the decrease in the height is faster than that in the middle width. Therefore, the oxidation of Au bipyramids is anisotropic. Figure 7J further shows the variation of the LSP wavelength versus the average aspect ratio. The increase in the LSP wavelength with increasing aspect ratio is nonlinear, which is similar to the case for the Au bipyramids shown in Figure 4H. However, the overall shapes of the two curves are different, because Au bipyramids are transformed into ellipsoids during oxidation.

Gold bipyramids exhibit a penta-twinned crystal structure.^[15,16] Each bipyramid has five domains with {111} twinning planes. We performed high-resolution (HR) TEM

imaging to characterize the crystal structures of the products obtained from oxidation. Figure 8A and B show representative HRTEM images for a slightly shortened bipyramid and an ellipsoid with smooth surfaces, respectively. Fringes running parallel to the longitudinal direction on both edges are observed for both nanostructures. These fringes become modulated in the central region into wider stripes. Such features are typical for a penta-twinned face-centered-cubic crystal structure, suggesting that the penta-twinned crystal structure is preserved throughout oxidation.

We lastly performed oxidation experiments on Au nanorods that were grown in CTBAB and CTAB solutions in order to investigate the effects of the nanoparticle shape and the stabilizing agent on the oxidation behavior. In addition, the oxidation of Au nanorods using H_2O_2 as an oxidizing agent has not been explored before. CTBAB- and CTAB-stabilized Au nanorods used for oxidation in our experiments were grown following the procedures reported previously.^[17,24] Figure 9A and B show the time-dependent extinction spectra acquired during the oxidation of CTBAB- and CTAB-stabilized Au nanorods, respectively.

The volume ratios of H_2O_2 to as-grown nanorod solutions are 0.5 for both oxidation experiments. The oxidation rates estimated by plotting the LSP wavelength versus the oxidation time are 1.4 and 15.3 nm min^{-1} , respectively (Table 2). Moreover, when the volume ratios of H_2O_2 to the nanorod solutions are 1.0, the obtained oxidation rates are 3.2 and 40 nm min^{-1} for CTBAB- and CTAB-stabilized nanorods, respectively (Table 2). Clearly, with the same amount of H_2O_2 , the oxidation of Au nanorods is several times faster than that of bipyramids in CTBAB solutions, and the oxidation of nanorods in CTAB solutions is more than 10 times faster than that in CTBAB solutions, although the concentration of CTBAB is one tenth that of CTAB. The slower oxidation of Au nanorods in CTBAB solutions further substantiates our previ-

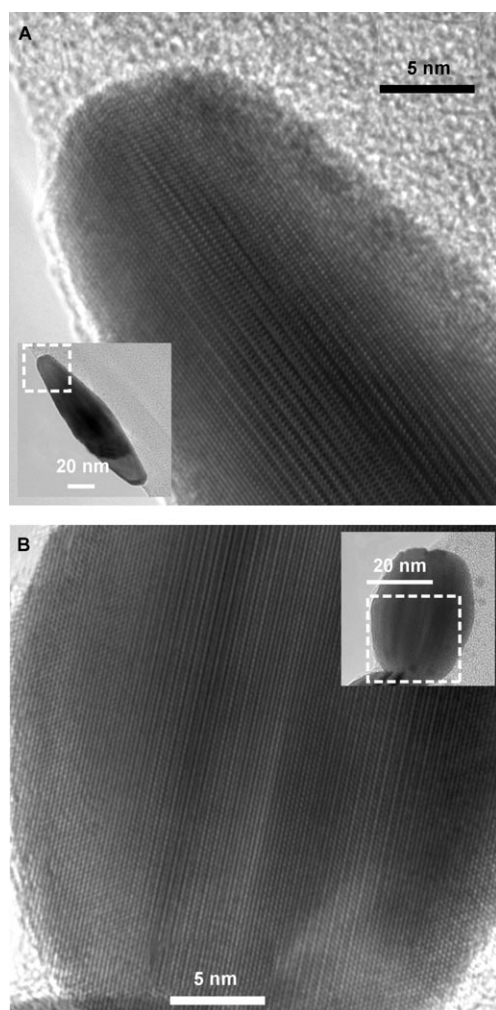


Figure 8. HRTEM images of the products obtained from oxidation. The samples shown in (A) and (B) underwent oxidation for 6 h and 12 h, respectively. The oxidation conditions are the same as those for the samples shown in Figure 7. The insets are TEM images of the corresponding oxidized bipyramids. The boxes indicate the locations where HRTEM images were taken.

ous assertion that the binding of CTBAB to Au surfaces is stronger than that of CTAB to Au surfaces.^[17] The slower oxidation of Au bipyramids than that of Au nanorods in the same surfactant solutions can be ascribed to the larger sizes of bipyramids. In our experiments, the volumes of CTBAB-stabilized Au bipyramids are 5–10 times larger than those of CTBAB-stabilized Au nanorods. Therefore it takes longer to induce an equal blue-shift in the LSP wavelength through oxidation for the bipyramids than for the nanorods.

3. Conclusions

The LSP properties of Au bipyramids and nanorods with similar LSP wavelengths have been investigated using the FDTD method. When excited at their LSP wavelengths, Au bipyramids exhibit a maximal electric-field-intensity en-

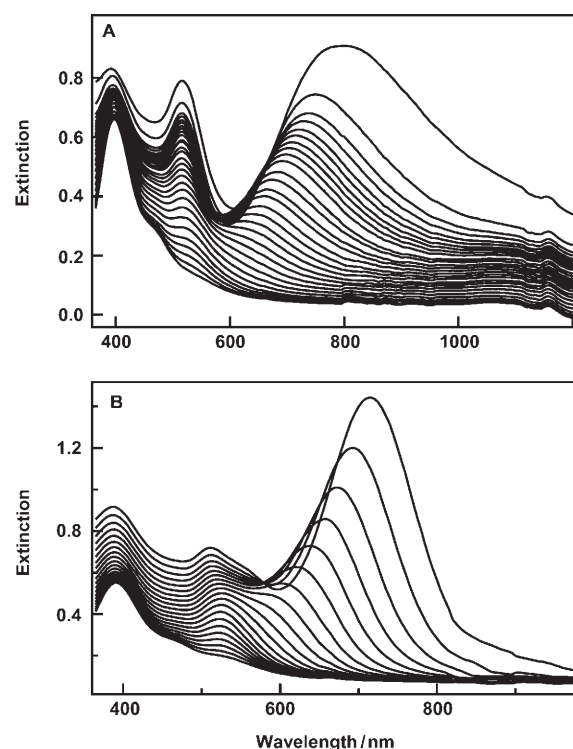


Figure 9. A, B) Time-dependent extinction spectra acquired for the oxidation of Au nanorods in CTBAB (0.01 M) and CTAB (0.1 M) solutions, respectively. The volume ratio of H₂O₂ (35 wt.%) to as-grown Au nanorod solutions is 0.5 for both oxidation experiments. The first four spectra in (A) were taken every 1.25 min, the next four spectra were taken every 3 min, and the remaining spectra were taken every 10 min. The spectra in (B) were acquired every 1.25 min.

hancement that is 3–6 times that of Au nanorods. The field-intensity enhancement associated with the LSP resonance is mainly determined by the particle shape instead of the particle volume. Encouraged by this result, we have used CTBAB as a stabilizing agent to prepare Au bipyramids with a seed-mediated growth method. The yields of the resulting bipyramids are 50–60% and they remain nearly unchanged when the LSP wavelengths of the bipyramids are tailored from 800 to 1300 nm by varying the seed amount. Oxidation experiments have further been carried out on CTBAB-stabilized Au bipyramids using H₂O₂ as the oxidizing agent. It has been found that oxidation preferentially occurs at surface sites with high curvature. The oxidation rate of Au bipyramids can be controlled by varying the amount of H₂O₂ and HCl. It has also been found that with the same amount of H₂O₂, the oxidation rate increases in the order of CTBAB-stabilized Au bipyramids, CTBAB-stabilized Au nanorods, and CTAB-stabilized Au nanorods. This oxidation approach can be used together with the seed-mediated growth in CTBAB solutions to produce Au bipyramids with LSP wavelengths finely controlled from 650 to 1300 nm.

4. Experimental Section

FDTD calculation: A software package, FDTD Solutions, developed by Lumerical Solutions, Inc., was employed to carry out FDTD calculations on Au bipyramids and nanorods. The gold dielectric function was represented using a Drude model,^[18] with parameters chosen to match bulk dielectric data measured previously.^[29] The refractive index of the surrounding medium, water, was taken to be 1.33. An electromagnetic pulse was launched in a box around a target Au nanostructure to simulate a propagating plane wave interacting with the nanostructure. Only the LSP resonance was calculated by setting the electric field of the plane wave parallel to the longitudinal direction. The nanorod was modeled as a cylinder capped with a half sphere at each end. The apexes of the bipyramid were modeled as spherical caps with five cut facets. Their sizes were taken to match the average values measured from TEM images as closely as possible. Different mesh sizes were used for each nanostructure. The changes of various parameters, including the plasmon wavelength, extinction cross section, line width, and field-intensity enhancement, as a function of the mesh size were obtained. For small mesh sizes, graded meshes were employed, where the size of the mesh cells on the target nanostructure was smaller than that in the surrounding region. Such nonuniform meshes can make FDTD calculation more accurate, while requiring less memory and less computation time than comparable uniform meshes.

Growth of gold bipyramids: Gold bipyramids were grown using the seed-mediated method in aqueous cationic surfactant solutions in the presence of AgNO₃. CTEAB, CTPAB, and CTBAB were synthesized by refluxing stoichiometric amounts of 1-bromohexadecane and corresponding trialkylamine in dry acetonitrile, as reported previously.^[17] Citrate-stabilized Au nanoparticles were used as seeds. Their preparation followed the previously reported procedure.^[15,16] Briefly, aqueous solutions of HAuCl₄ (0.01 M, 0.125 mL) and sodium citrate (0.01 M, 0.25 mL) were first added into deionized water (9.625 mL), and then a freshly prepared, ice-cold solution of NaBH₄ (0.01 M, 0.15 mL) was added under vigorous stirring. The resulting seed solution was kept for at least 2 h before use. For a typical preparation of Au bipyramids in CTBAB solutions, a growth solution was first prepared by the addition of aqueous solutions of HAuCl₄ (0.01 M, 0.3 mL), ascorbic acid (0.1 M, 0.1 mL), and AgNO₃ (0.01 M, 0.45 mL) into an aqueous CTBAB solution (0.01 M, 7.125 mL). The citrate-stabilized seed solution (0.075 mL) was then added. The reaction solution was mixed by gentle inversion for 10 s and then left undisturbed overnight in an isothermal oven at 65 °C. Similar preparations with the use of different volumes of the ascorbic acid and seed solutions were carried out in order to investigate the effects of the amounts of ascorbic acid and the seeds on the growth of Au bipyramids in CTBAB solutions. This preparation procedure was also followed to grow Au bipyramids in CTAB, CTEAB, and CTPAB solutions, respectively, where the concentrations of the three surfactants were 0.1 M and the growth was at room temperature.

Growth of gold nanorods: CTAB-stabilized Au nanorods were grown as reported previously,^[24,30] and CTBAB-stabilized Au

nanorods were prepared according to our previously reported procedure.^[17]

Oxidation of gold bipyramids and nanorods: As-grown Au bipyramid and nanorod solutions were used for oxidation. 35 wt.% H₂O₂ was used as the oxidizing agent. Oxidation was carried out by the addition of the H₂O₂ solution into Au bipyramid and nanorod solutions. The oxidation process was monitored by taking extinction spectra as a function time. The volume of the H₂O₂ solution was varied and different volumes of HCl (1 M) was added in order to control the oxidation rate.

Characterization: Extinction spectra were taken on a Hitachi U-3501 UV/Vis/NIR spectrophotometer. Rectangular quartz cuvettes with an optical path length of 1 cm were used. Low-magnification TEM images were acquired using a FEI CM120 microscope at 120 kV, and they were used to determine the yields and sizes of Au nanostructures. HRTEM images were recorded using a TECNAI 20 ST microscope at 200 kV. For TEM characterization, as-grown and oxidized Au bipyramid solutions (6 mL each) were centrifuged at 12 000 *g* for 6 min. The precipitates were redispersed in deionized water (6 mL each), centrifuged again at 12 000 × *g* for 6 min, and finally redispersed in deionized water (0.4 mL each). 0.01 mL of each resulting bipyramid solution was drop-cast onto a lacey-formvar carbon-stabilized TEM grid and allowed to dry in air overnight.

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